

Yifei (Diana) Tao

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. in Computer Science; Minor in Data Science

Aug 2022 – May 2026

- Relevant Coursework: Algorithms, Data Structures, Database Systems, Machine Learning, Computer Vision, Optimization Models, Computer Security, Probability & Statistics, Machine Structures

EXPERIENCE

AI Engineer

May 2026 – Present

Prompt Driven

Palo Alto, CA

- Built **AI-assisted developer tooling** for automated code review, requirement validation, interactive repair, structured reporting, and hosted **GitHub App** execution.
- Improved cloud **AI-agent reliability** with telemetry, retry/resume logic, provider failure classification, checkpoint handling, shared auth cooldowns, and regression tests.
- Debugged platform issues across **GitHub App** workers, Firebase/GitHub auth, Firestore state, CORS configs, staging deployments, and **Terraform-managed GCP/Firebase** test environments.

Machine Learning Engineering Intern

Jan 2026 – May 2026

C. Light Technologies

Santa Clara, CA

- Fine-tuned **RETFound Vision Transformer** models on retinal imaging datasets to predict neurological disease severity (EDSS) in medical imaging classification and regression tasks.
- Developed spatiotemporal representation learning pipelines using **VideoMAE** and **U-Net** architectures for retinal video analysis across clinical datasets.
- Built scalable GPU training workflows on **AWS EC2**, including frame extraction, dataset preprocessing, and automated model evaluation pipelines.
- Implemented reproducible ML experimentation pipelines to accelerate model iteration and large-scale training experiments.

Software Engineering Intern

Sep 2025 – Dec 2025

C. Light Technologies

Santa Clara, CA

- Designed and implemented an **agent orchestration system** coordinating LLM agents for retinal data analysis workflows and internal research tooling.
- Architected a **Django-based backend platform** providing APIs and service orchestration across **AWS S3** and SQL databases.
- Built a **Redis caching layer** to support multi-step backend workflows, improving API responsiveness and reliability.

Research Software Developer

Sep 2025 – May 2026

UCSF TECH Lab — SocialPlai

San Francisco, CA

- Built **full-stack** features for SocialPlai, a UCSF TECH Lab web platform hosting cognitive and social games for behavioral health research.
- Implemented **session-management** and data-capture workflows to record participant responses, mouse movements, and real-time interaction events.
- Collaborated with research stakeholders to translate neuropsychological study requirements into scalable software features supporting solo and multiplayer assessments.

Undergraduate Research Assistant

Sep 2024 – May 2025

Center for Data Science & Environment, UC Berkeley

Berkeley, CA

- Fine-tuned audio classification models (**BirdNET**, **Google Perch**, **ResNet**) on **10,000+ annotated audio samples** for biodiversity monitoring.
- Improved species detection accuracy by modifying model architectures and evaluating performance using **ROC-AUC** and **precision-recall** metrics.
- Developed annotation and visualization tools including a web-based labeling interface and model evaluation dashboards.

TECHNICAL SKILLS

Languages: Python, Java, C, Go, SQL, JavaScript, TypeScript, R

Frameworks: React, Node.js, Express.js, Django

Machine Learning: TensorFlow, PyTorch, Scikit-learn, Keras

Tools & Platforms: AWS, Google Cloud Platform, Redis, Git, Docker, VS Code, IntelliJ

Libraries: pandas, NumPy, Matplotlib